

Abstract

Automated Diabetic Retinopathy (DR) detection, screening and diagnosis are critical to save vision loss of patients and assist ophthalmologists in mass screening. DR screening aims at early treatment by detecting the disease before progression. Current DR analysis systems use digital fundus images to reduce the high cost and subjectivity of manual grading. Researchers continue to develop automated screening systems to improve consistency and reduce clinician burden. However, class imbalance and data scarcity remain significant challenges for robust deep learning-based diagnosis.

The proposed system introduces a hybrid Generative Adversarial Network and Convolutional Neural Network (GAN-CNN) framework for DR detection and severity grading. The approach utilizes a Conditional GAN (cGAN) to generate synthetic fundus images that address class imbalance across five severity grades. A CNN architecture, based on ResNet-50, is then trained on the augmented dataset to perform multi-class classification. The system automates preprocessing, feature learning, and classification without requiring manual lesion extraction.

The hybrid GAN-CNN model achieves an accuracy of 91.2% with a cross-entropy loss of 0.48, demonstrating a 15% improvement over state-of-the-art methods. The framework significantly enhances performance on minority classes while maintaining computational efficiency, making it suitable for real-world clinical deployment. This approach shows strong potential for scalable, accurate, and automated DR screening in diverse healthcare settings.

Index Terms— diabetic retinopathy; deep learning; convolutional neural networks; generative adversarial networks; medical image synthesis; computer-aided diagnosis; severity grading; fundus imaging.